

Welcome to the Course

Using Object Detection Models

- ✔ **Video:** Introduction to Object Detection with CNNs  
5 min
- ✔ **Video:** Using Pre-trained Object Detectors  
2 min
- ✔ **Reading:** Installing Pre-trained Object Detectors  
10 min
- 📖 **Reading:** Using Detection Models on Images and Videos  
30 min
- 📖 **Reading:** YOLO Detectors in MATLAB Reference  
5 min
- 📝 **Quiz:** Getting Started with Object Detection  
30 min

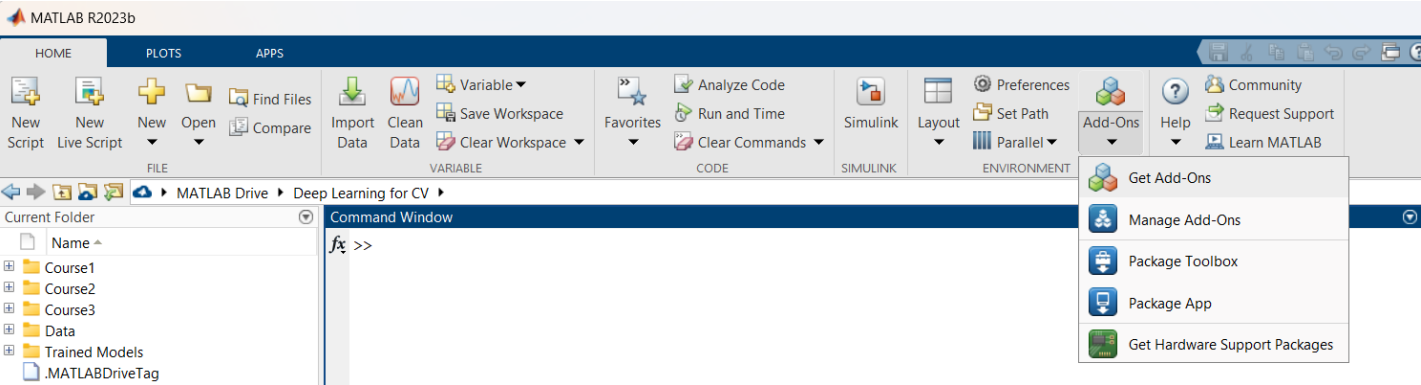
# Installing Pre-trained Object Detectors

To follow along and do the exercises in this course, you need to install two Add-On Packages to MATLAB. The Add-Ons will install two different YOLO models that will use throughout the course.

Follow the instructions below to install them.

## Install the YOLOv4 pre-trained model

- Go to the Add-On Explorer in the MATLAB Toolstrip and select "Get Add-Ons."



- Search for "**computer vision for yolo v4.**" The first search result should be the YOLOv4 pre-trained model by MathWorks. (You will only see the "Installed" banner if you've already installed the add-on)

Installed

### Computer Vision Toolbox Model for YOLO v4 Object Detection

by MathWorks Computer Vision Toolbox Team **STAFF**

Detect objects using a pretrained **YOLO v4** object detection network trained on COCO dataset.

objects from the COCO dataset. Refer to the **YOLO v4** documentation for more information.

MathWorks Optional Feature

- Select the add-on shown above and follow the instructions to install the model.

## Install the Computer Vision Toolbox Automated Visual Inspection Library

This add-on includes a new YOLOX detector that you will use.

- Go to the Add-On Explorer in the MATLAB Toolstrip and select "Get Add-Ons" as you did for YOLO v4
- Search for "Computer Vision Visual Inspection Library"

Installed

### Computer Vision Toolbox Automated Visual Inspection Library

by MathWorks Computer Vision Toolbox Team **STAFF**

Identify anomalies or defects in images to assist and improve quality assurance processes.

The **Computer Vision** Toolbox™ Automated **Visual Inspection** Library offers functions that enable you to train, evaluate, and deploy anomaly detection and object detection networks for use in **visual**

MathWorks Optional Feature

- Select the add-on shown above and follow the instructions to install the model.

## Additional Models are Available on the Model Hub

Be sure to check out the [MATLAB Model Hub](#) [↗](#) after the course. New models are regularly added to help you stay up to date.

Go to next item      ✓ Completed

